

STE Mapbook Workflow — User Guide

This guide walks you through configuring and running the STE Mapbook Workflow, which generates a multi-section mapbook report for wildlife tracking data sourced from EarthRanger and Google Earth Engine.

Overview

The workflow produces, for each tracked subject:

- Six interactive map visualizations (movement tracks, speed, day/night, home range, mean speed raster, seasonal home range)
 - A Word document mapbook (`.docx`) with a cover page and one section per subject
 - An interactive widget dashboard
-

Prerequisites

Before running the workflow, ensure you have:

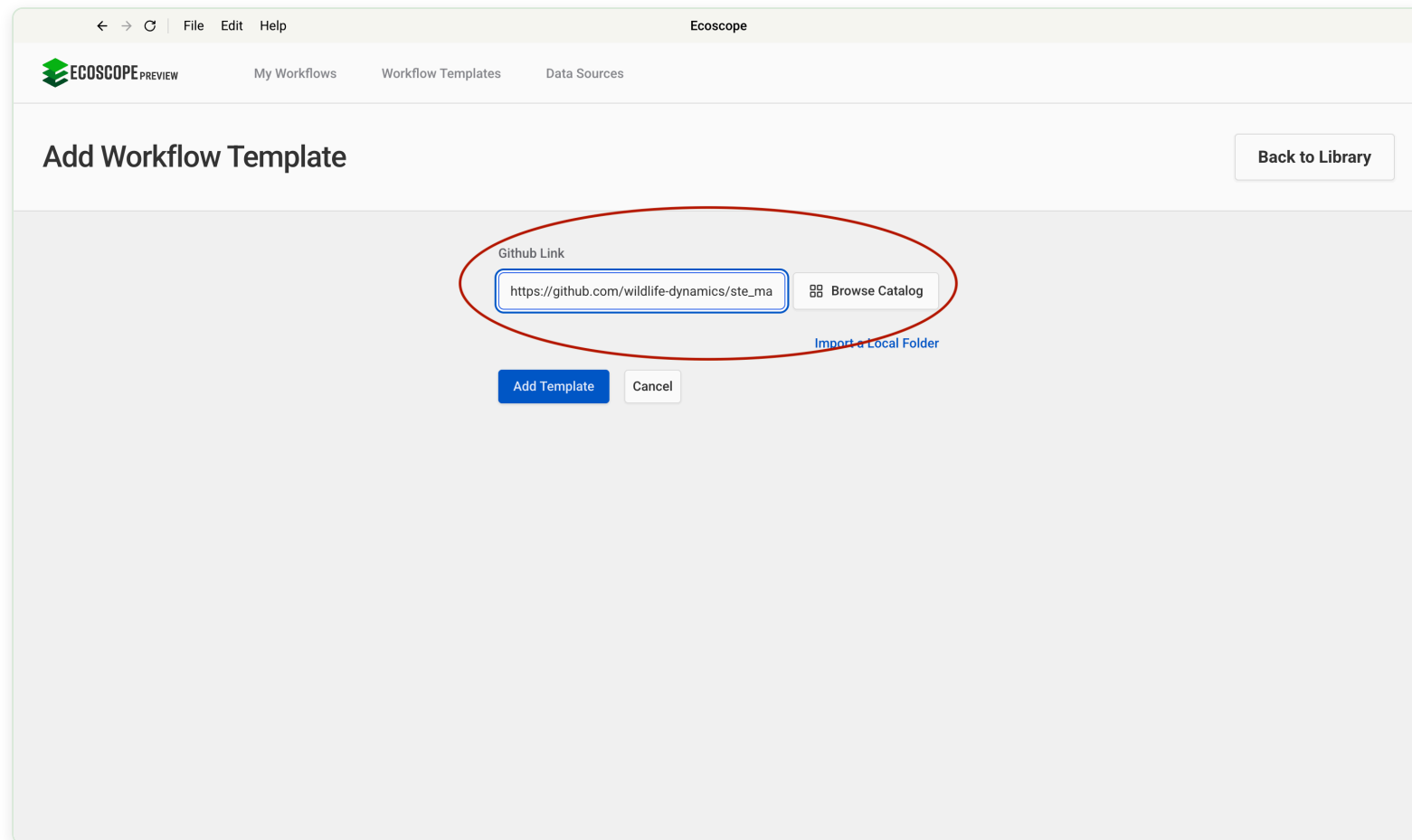
- Access to an **EarthRanger** instance with a configured data source
 - Access to a **Google Earth Engine** project
 - The `landDx` geodatabase (downloaded automatically from Dropbox by default — no local copy required)
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Step-by-Step Configuration

Step 1 — Add the Workflow Template

In the workflow runner, add a new template using the following GitHub repository URL:

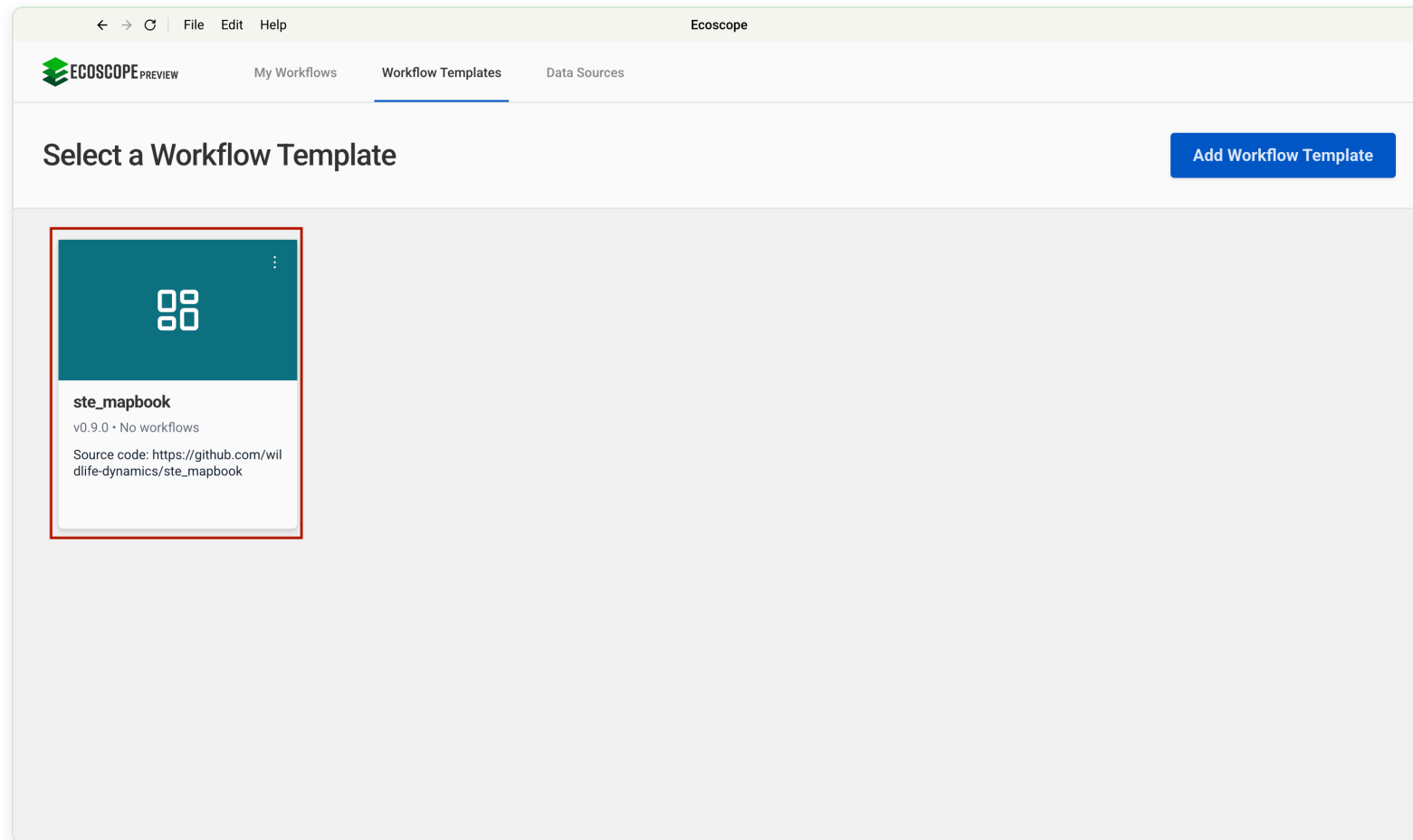
```
https://github.com/wildlife-dynamics/ste_mapbook.git
```



Add Template

Step 2 — Select the Template

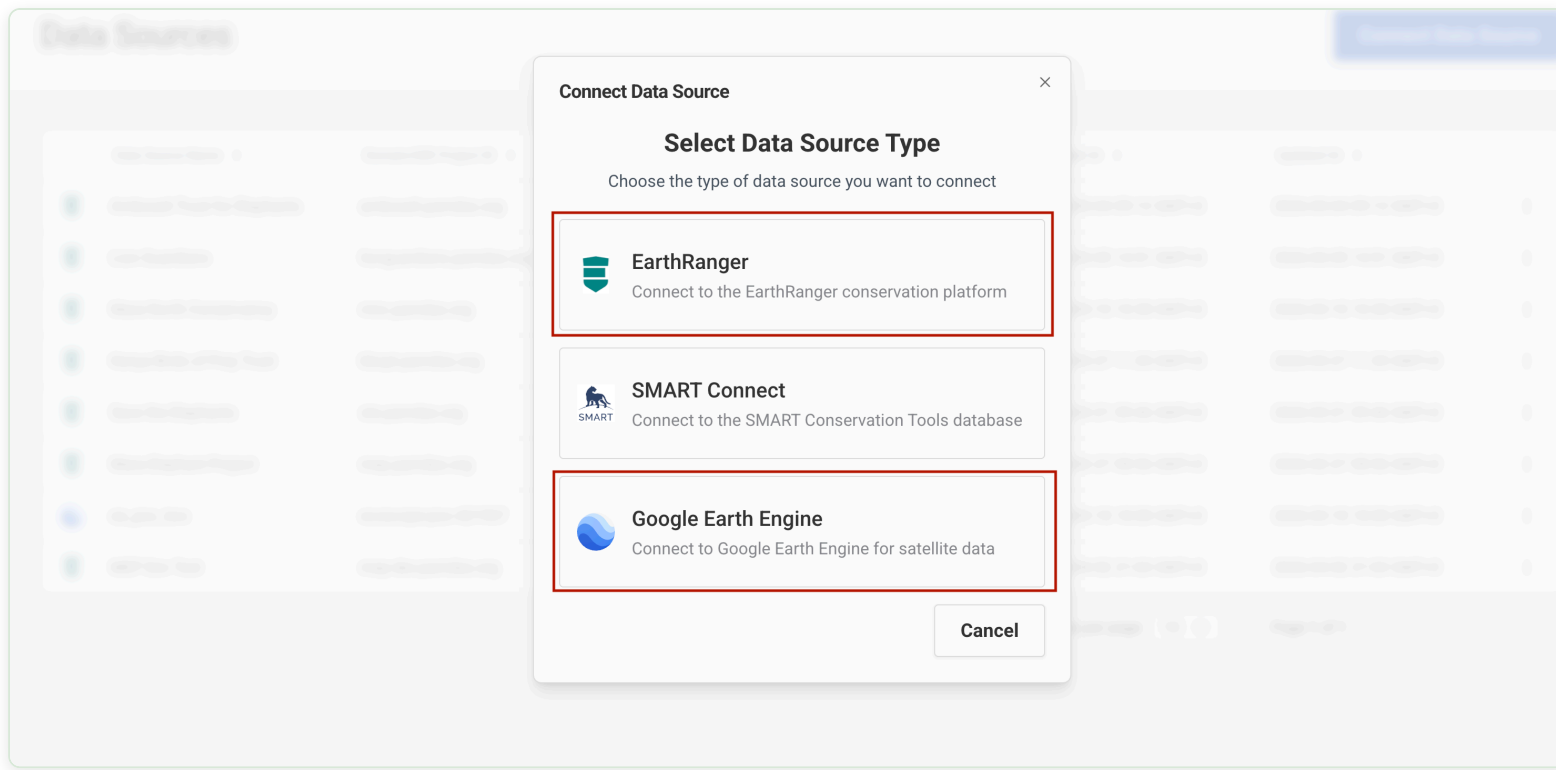
After adding the template, select it from the available templates list to load the STE Mapbook workflow.



Select Template

Step 3 — Configure Connections

Both an **EarthRanger** connection and a **Google Earth Engine** connection are required for the workflow to run. Configure these before proceeding to the workflow parameters.



Configure Connection

Configure EarthRanger Connection

Enter your EarthRanger server URL and authentication credentials to establish the EarthRanger data connection.

EARTH RANGER
A product of AI2

Connect Ecoscope to EarthRanger

Data Source Name

EarthRanger URL

EarthRanger Username

EarthRanger Password

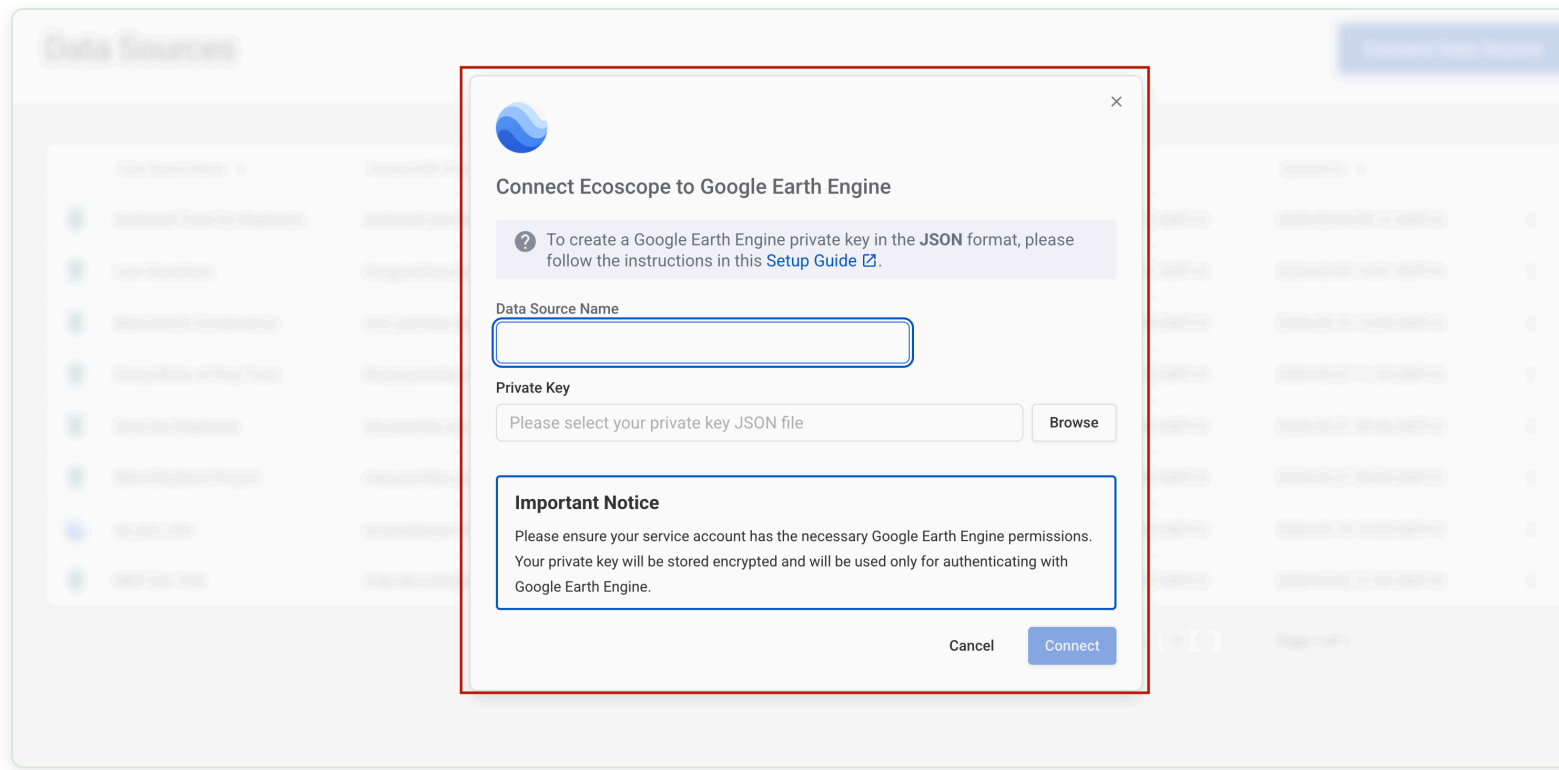
⚠ Important Notice
Please double-check your credentials carefully. Credentials entered are NOT validated during setup. Any authentication errors will only show up when running a workflow.

Cancel **Connect**

EarthRanger Connection

Configure Google Earth Engine Connection

Enter your Google Earth Engine project name to establish the GEE connection used for seasonal analysis.



Earth Engine Connection

Step 4 — Set the Workflow Time Range

Define the analysis period by setting the **Since** (start) and **Until** (end) dates. All movement data, trajectories, and home ranges will be computed within this window.

← → ↻ | File Edit Help Ecoscope

Workflow Templates > Workflow Configuration

Set workflow details

Define analysis time range

Set previous period range

Connect to earth ranger

Connect to earth engine

Subject Group

Load landDx database

Trajectory Segment Filter

Zoom to gdf extent

Report logo

Set workflow details

Add information that will help to differentiate this workflow from another.

Workflow Name*

Workflow Description

Define analysis time range

Choose the period of time to analyze.

Timezone *

Since *

Until *

The start time

The end time

Set previous period range

Define the previous period for comparison. By default, it will match the current period, but you can choose another option like "Previous month" or "Previous year", or specify a custom range.

Cancel Submit

Workflow Time Configuration

Step 5 — Set the Previous Period and Connection

Select a **previous period** to use as a comparison baseline on the Movement Tracks map, and confirm the active data connections.

Option	Description
Same as current period	Mirrors the current period length, ending at the current start date
Previous month	Calendar month before the current period
Previous 3 months	Three calendar months before the current period
Previous 6 months	Six calendar months before the current period
Previous year	One calendar year before the current period
Enter start date	Manually specify a start date for the previous period

The screenshot shows the Ecoscope application interface. On the left is a sidebar with a list of steps: 'Set workflow details', 'Define analysis time range', 'Set previous period range', 'Connect to earth ranger', 'Connect to earth engine', 'Subject Group', 'Load landDx database', 'Trajectory Segment Filter', 'Zoom to gdf extent', and 'Report logo'. The main area contains three highlighted sections:

- Set previous period range**: A section with a description: 'Define the previous period for comparison. By default, it will match the current period, but you can choose predefined options like "Previous month" or "Previous year", or specify a custom range.' It includes a dropdown menu labeled 'Option*' with the text 'Choose which period you'd like to select your date from'. Below this is another dropdown menu labeled 'PreviousPeriodType*' with the text 'Same as current period' and a subtext 'Select the previous period type'.
- Connect to earth ranger**: A section with the instruction 'Select one of your configured data sources.' It features a dropdown menu labeled 'Data Source*' with the value 'Amboseli Trust for Elephants'.
- Connect to earth engine**: A section with the instruction 'Select one of your configured data sources.' It features a text input field labeled 'Data Source*' containing the value 'ee_poc_two'.

At the bottom of the main area, there is a 'Subject Group' section with a description 'Choose subject group to analyze' and a text input field. To the right of the main area are 'Cancel' and 'Submit' buttons.

Previous Period and Connection

Step 6 — Select Subject Group and Load landDx Database

Enter the **Subject Group Name** exactly as it appears in EarthRanger (case-sensitive). The default value is `Elephants`.

For the **landDx geodatabase**, which contains protected area boundaries used as base layers on all maps:

- If you do not have the database stored locally, a **Dropbox download URL is provided by default** — no action is needed.
- If you have a local copy, you can supply the path to your `.gpkg` file instead.

The screenshot shows the Ecoscope web application interface. On the left is a sidebar with navigation links: 'Set workflow details', 'Define analysis time range', 'Set previous period range', 'Connect to earth ranger', 'Connect to earth engine', 'Subject Group', 'Load landDx database', 'Trajectory Segment Filter', 'Zoom to gdf extent', and 'Report logo'. The main content area has two sections highlighted with red boxes. The top section, 'Subject Group', contains the instruction 'Choose subject group to analyze', a label 'Subject Group Name*', a text input field containing 'Elephants', and a note 'Enter the subject group name exactly as it appears in EarthRanger (case-sensitive)'. The bottom section, 'Load landDx database', contains the instruction 'Load the landDx geodatabase and prepare it for mapping. The geodatabase will be filtered to include only protected areas (Community Conservancies, National Reserves, National Parks) and styled for visualization.', a label 'Input Method*', a dropdown menu with 'Download from URL' selected, and a label 'URL*' with a text input field containing a long Dropbox URL. Below the URL field is a small text label 'URL to download a file'.

Subject Group and landDx Database

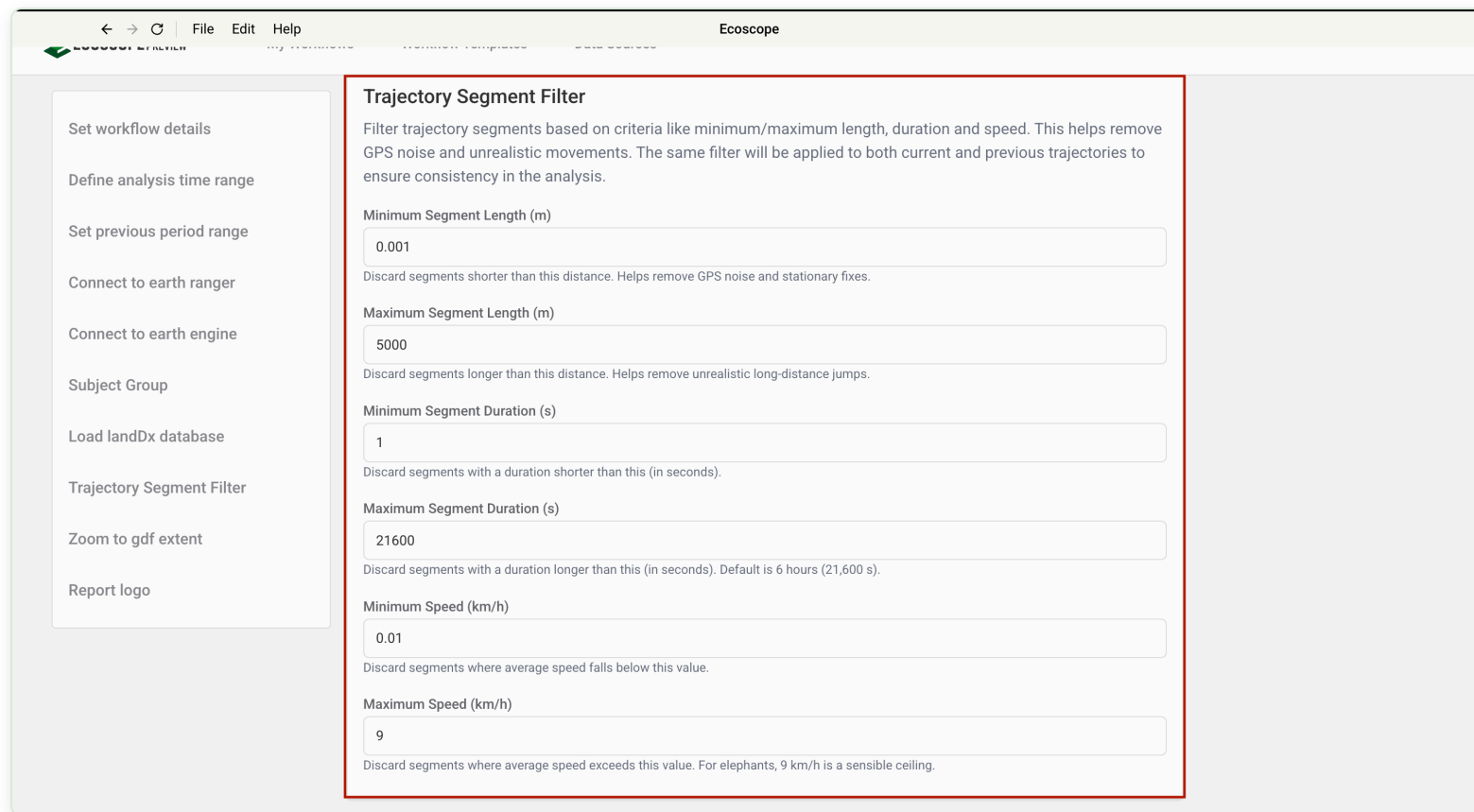
Step 7 — Configure Trajectory Segment Filter

These filters remove GPS noise and unrealistic movements before trajectory analysis. The same filter is applied to both the current and previous period trajectories.

Field	Default	Description
Minimum Segment Length (m)	0.001	Discard segments shorter than this distance

Field	Default	Description
Maximum Segment Length (m)	5000	Discard segments longer than this distance
Minimum Segment Duration (s)	1	Discard segments shorter than this duration
Maximum Segment Duration (s)	21600	Discard segments longer than this duration (6 hours)
Minimum Speed (km/h)	0.01	Discard segments below this average speed
Maximum Speed (km/h)	9	Discard segments above this average speed

Adjust these values to suit the movement characteristics of your study species.



Configure Trajectory Segment Filter

Step 8 — Zoom to GDF Extent and Report Logo

Zoom to GDF Extent

Set the **Expansion Factor** to control the map padding around the data boundary. A value of `1.0` gives a tight fit; `1.05` (default) adds 5% padding; `1.2` adds 20% padding.

Report Logo

The logo appears on the mapbook cover page. You can either:

- Provide a **URL** to a PNG or JPG image to download it automatically.
- Provide a **local file path** to an image on your machine.

The screenshot shows a web interface with a sidebar on the left containing a list of configuration options: 'Set previous period range', 'Connect to earth ranger', 'Connect to earth engine', 'Subject Group', 'Load landDx database', 'Trajectory Segment Filter', 'Zoom to gdf extent', and 'Report logo'. The main content area has two red-bordered boxes. The top box, titled 'Zoom to gdf extent', contains a description: 'Calculate the geographic extent of the movement tracks and determine an appropriate zoom level and center point for the map. This ensures that all tracks are visible and well-framed when the map is rendered.' Below this is an 'Expansion Factor' input field with the value '1.05' and a help text: 'Adds padding around the map boundary. 1.0 = tight fit, 1.05 = 5% padding (default), 1.2 = 20% padding.' The bottom box, titled 'Report logo', contains a description: 'Download or fetch the logo to be used in the report cover page. This task can be customized to fetch logos from different sources based on user requirements.' Below this is an 'Input Method*' dropdown menu with 'Download from URL' selected. Underneath is a 'Download from URL' section with a 'URL*' input field and a help text: 'URL to download a file' and 'Provide the organization logo to display on the report cover page (PNG or JPG recommended)'. At the bottom right of the interface are 'Cancel' and 'Submit' buttons.

Zoom to GDF Extent and Report Logo

Running the Workflow

Once all parameters are configured, submit the workflow. The runner will:

1. Pull movement data from EarthRanger for the specified subject group and time range.
2. Filter trajectories using the segment filter settings.
3. Compute home ranges, speed rasters, and seasonal ranges using Google Earth Engine.
4. Generate all map visualizations and the Word mapbook.

5. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS` .

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/` :

File	Description
<code>trajectories.geoparquet</code>	Current period trajectories
<code>previous_period_trajectories.geoparquet</code>	Previous period trajectories
<code>relocations.geoparquet</code>	Current period relocations
<code>previous_period_relocations.geoparquet</code>	Previous period relocations
<code>*.geoparquet</code> (ETD, MCP, seasonal)	Home range polygons per subject
<code>*_movement_tracks.html</code>	Interactive movement tracks map per subject
<code>*_speedmap.html</code>	Interactive speed map per subject
<code>*_day_night.html</code>	Interactive day/night map per subject
<code>*_homerange.html</code>	Interactive home range map per subject
<code>*_mean_speed_raster.html</code>	Interactive mean speed raster per subject
<code>*_seasonal_home_range.html</code>	Interactive seasonal home range per subject
<code>mapbook_context_page.docx</code>	Cover page document

File	Description
*.docx (per subject)	Individual subject report sections
Merged mapbook .docx	Final combined Word report